

Predicting Molecular Trajectories of a Nanomotor with
Neural Networks
WHK Research Project

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Presentation outline

1. Fundamentals
2. Scope of this talk
3. Training the network
4. Results and Outlook

Fundamentals

Simple Example of a Neural Network

- ▶ A **fully connected neural network** maps an input vector $\mathbf{x}$ to an output $\mathbf{y}$, e.g. $f : \mathbb{R}^3 \rightarrow \mathbb{R}^2$, $\mathbf{x} \mapsto \mathbf{y}$
- ▶ Each neuron computes a **weighted sum** followed by a **nonlinear activation**:
$$h_2 = \sigma(x_1 w_{12} + x_2 w_{22} + x_3 w_{32})$$
- ▶ Stacking layers yields the overall mapping: $f(\mathbf{x}) = \sigma(\mathbf{W}_{32} \sigma(\mathbf{W}_{21} \mathbf{x}))$

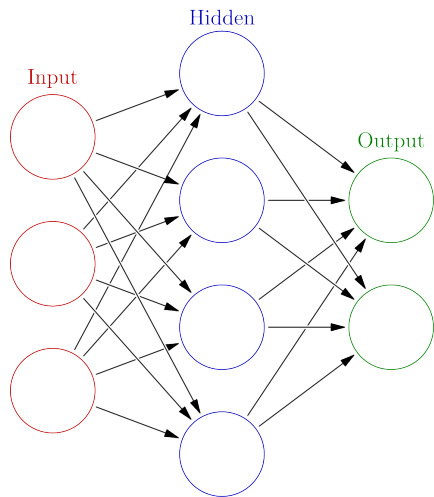


Figure: Fully connected neural network, adapted from [1]

Molecular Dynamics (MD)

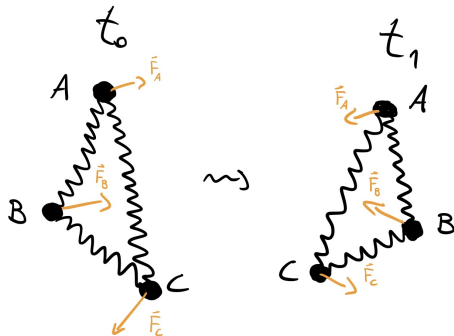


Figure: Simple sketch of one MD step for point masses connected via springs.

- ▶ **Goal:** Predict atomic trajectories by integrating Newton's equations of motion.
- ▶ Requires an algorithm to compute the **forces** on atoms from their current positions.
- ▶ Possible **force predictors**:
 - ▶ Density Functional Theory (DFT)
 - ▶ Extended Tight Binding (XTB)
 - ▶ Neural Networks (NN)

What is a Nanomotor?

- ▶ Wikipedia: *A nanomotor is a molecular or nanoscale device capable of **converting energy into movement**.*
- ▶ In this work, we study the molecule $C_{27}H_{20}O$ (CPNX), which can act as a **molecular propeller**.

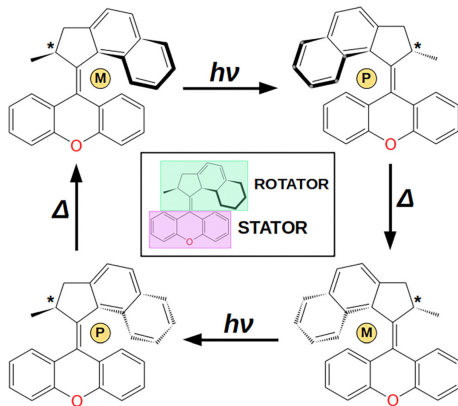
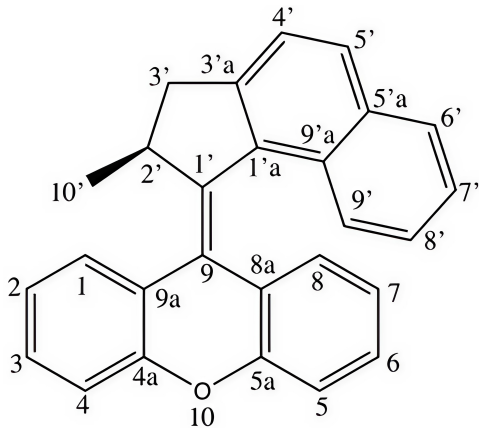


Figure: Schematic representation of nanomotor states, adapted from [2, p. 2]

Scope of today's talk



- ▶ Work on one dataset, containing mostly **one conformation**
- ▶ Prediction of **energy** and **forces** given atomic positions
- ▶ Molecule is only examined in **ground state**

Figure: Figure adapted from [2, p.13]

Training the network

Underlying Dataset

- ▶ Data sampled in **NVT ensemble** using XTB
- ▶ Split in train (12615), validation (3154) and test data (3943)
- ▶ **Center of mass** and **energy offset** subtracted

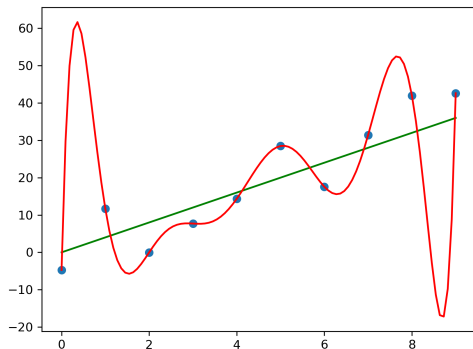
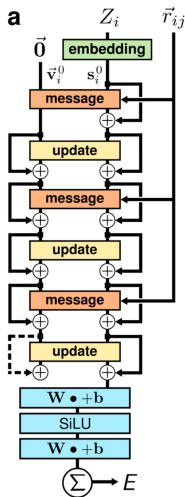


Figure: Example of overfitting taken from [6].

Notes on the Used Network



- ▶ The molecule is represented as a **graph**.
- ▶ Each atom corresponds to a **node** with both a scalar and a vector representation.
- ▶ The network predicts **energies** and computes **forces as their derivatives**.
- ▶ Energies are predicted **atomwise** and summed to obtain the total energy.

Figure: Architecture of the employed network, adapted from [4].

Details on Training

- ▶ Training performed using **SchNetPack** [5]
- ▶ **Loss function:** Mean Squared Error (MSE)
- ▶ **Loss weighting:** energies (0.01) and forces (0.99)
- ▶ Training for **1000 epochs** takes roughly six hours but does not converge

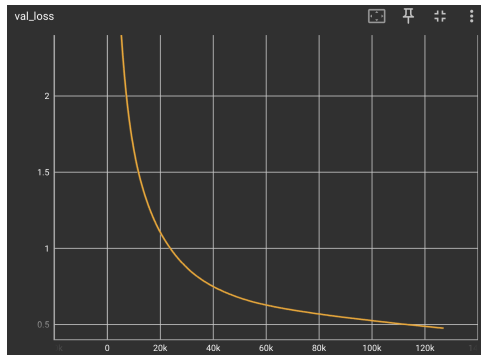


Figure: Validation loss as a function of training steps for 1000 epochs (units in Å and kcal/mol).

Impact of units

- ▶ Change of units (COU) **affects model performance** (see results section)
- ▶ Possible explanation: COU **affects weights** of energies and forces within the MSE
- ▶ Trying to counteract this effect, was not yet fruitful (see starred entry in results section)
- ▶ More on this later, if you are interested!

Results

Performance on Validation Data

Molecule	Training Units	Energy MAE [kcal/mol]	Forces MAE [kcal/mol/Å]
CPNX	Å, kcal/mol	0.1896 ± 0.0134	0.2999 ± 0.0110
CPNX	Å, eV	0.1581 ± 0.0027	0.3174 ± 0.0067
CPNX	Å, Hartree	0.2960 ± 0.0050	0.5154 ± 0.0100
CPNX	Å, Hartree*	0.3053 ± 0.0085	0.4993 ± 0.0119
<i>C₉O₈H₄</i>	Å, kcal/mol	0.11 – 0.38	0.14 – 1.17

Table: Mean absolute errors (MAEs) with corresponding standard errors on the validation data, for models trained in three different unit systems. The reference values for Acetylsalicylic (*C₉O₈H₄*) are taken from [5, p.453]. Note that the chemical accuracy is more or less defined as 1 kcal/mol

XTB vs. NN Trajectory

- ▶ Take last sample of test data as starting configuration
- ▶ Using its velocities should ensure that **trajectory is equilibrated**
- ▶ Predict trajectory with XTB and NN for **5000 steps à one femtosecond**
- ▶ Simulating this with NN takes around five minutes

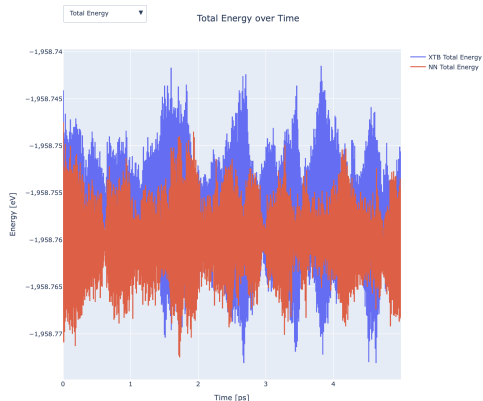


Figure: Plot of predicted energies from XTB and the neural network taken from [7].

Per-atom fluctuations

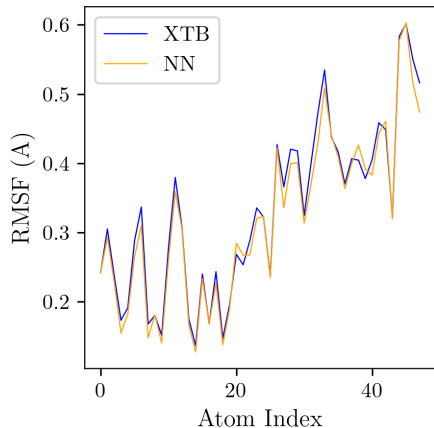


Figure: Average RMSF per atom.

- ▶ Take average over positions to compute **reference structure**
- ▶ Compute root mean squared **fluctuations** (RMSF) from reference structure **per atom over time**
- ▶ Indices $[0, \dots, 47] \hat{=} \text{atoms}$
 $[O, C, \dots H, \dots]$
- ▶ Pearson correlation of 0.9934

Outlook

- ▶ Train on all four conformations
- ▶ Try to use trained model as foundational model:
 - ▶ Finetune it for specific needs
 - ▶ Perform MD on the fly
- ▶ Move on to excited states
 - ▶ Predict non-adiabatic couplings
 - ▶ Employ surface hopping (e.g. fewest switches)
 - ▶ SHARC package [3]

Sources I

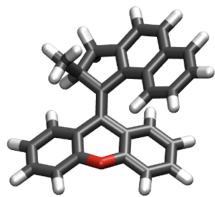
- [1] Glosser.ca. *Neural network (machine learning)*.
<https://commons.wikimedia.org/w/index.php?curid=24913461>. CC BY-SA 3.0. 2025.
- [2] Jesús Lucia-Tamudo, Michelle Menkel-Lantz, and Enrico Tapavicza. “First principles prediction of wavelength-dependent isomerization quantum yields of a second-generation molecular nanomotor”. In: *Physical Chemistry Chemical Physics* 27.23 (2025), pp. 12519–12531.
- [3] Sebastian Mai, Philipp Marquetand, and Leticia González. “Nonadiabatic dynamics: The SHARC approach”. In: *Wiley Interdisciplinary Reviews: Computational Molecular Science* 8.6 (2018), e1370.
- [4] Kristof Schütt, Oliver Unke, and Michael Gastegger. “Equivariant message passing for the prediction of tensorial properties and molecular spectra”. In: *International conference on machine learning*. PMLR. 2021, pp. 9377–9388.

Sources II

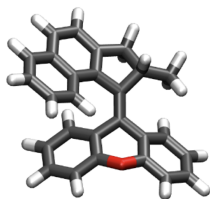
- [5] KT Schütt et al. “SchNetPack: A deep learning toolbox for atomistic systems”. In: *Journal of chemical theory and computation* 15.1 (2018), pp. 448–455.
- [6] ThirdOrderLogic. *Neural network (machine learning)*.
<https://commons.wikimedia.org/w/index.php?curid=150012280>. CC BY 4.0. 2025.
- [7] Ferdinand Tölkes. *Molecular Dynamics of a nanomotor using SchNetPack*.
<https://github.com/FerdinandToelkes/Molecular-Dynamics-with-Neural-Networks>. GitHub repository. 2025.

Backup Slides

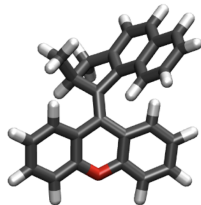
Dataset overview



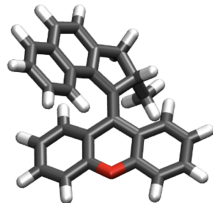
anti-M



anti-P



syn-M



syn-P

- ▶ **Four important conformations:**
anti-M, syn-M, anti-P and syn-P
- ▶ Can be roughly clustered by using **dihedral angles**, e.g.

$$\text{M: } \beta < 0, \quad \text{P: } \beta > 0$$

and furthermore

$$\begin{aligned} \text{syn-M: } \beta < 0 \quad \& \quad \delta < 0, \\ \text{anti-M: } \beta < 0 \quad \& \quad \delta > 0 \end{aligned}$$

Figure: Figure adapted from [2, p.11]

Dataset overview

- ▶ Data sampled in **NVT ensemble**
- ▶ Split in train (12615), validation (3154) and test data (3943)
- ▶ **Center of mass** and **energy offset** subtracted
- ▶ Configurations in T300_1 mostly in **anti-M**

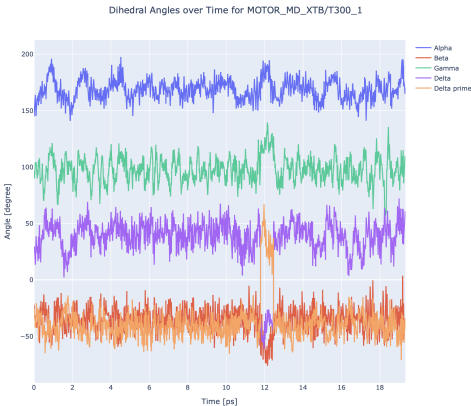


Figure: Plot of dihedral angles for T300_1 taken from [7].

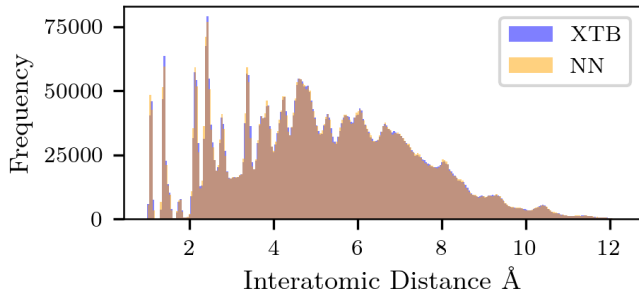
Correlation

- ▶ How similar are the trajectories around a certain point?
- ▶ Correlation as **measure of similarity**
- ▶ At step s take the next n steps to compute the **Pearson correlation**
- ▶ Repeat for each time step



Figure: Plot of correlations between predicted energies from XTB and the neural network taken from [7].

Interatomic distances



- ▶ Plot of distribution of interatomic distances
- ▶ $48 \cdot 47/2 = 1128$ distances per step
- ▶ Flattened into array of length $1128 \cdot \text{\#steps}$

Radius of gyration

- ▶ Radius of gyration is the distance from a systems **rotational axis** to its **center of mass**
- ▶ Describes **how masses** of system are **distributed**

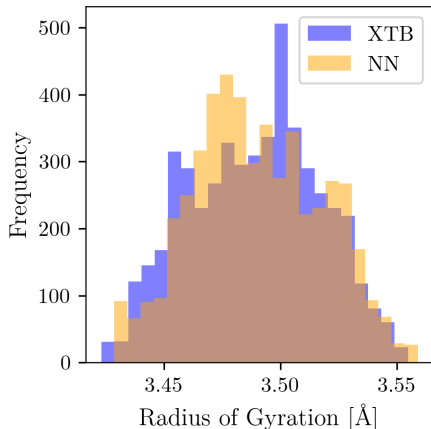
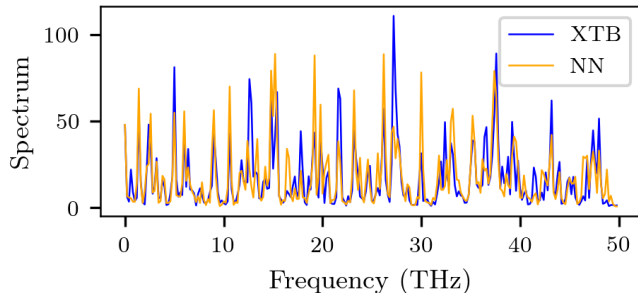


Figure: Distributions of the radius of gyration.

Power spectrum



- ▶ Excerpt of power spectra from XTB and NN trajectories, $r_{\text{Pearson}} = 0.7125$
- ▶ Calculated as Fourier transform of the velocity autocorrelation function
- ▶ Gives insights into active vibrational modes